

CALGARY CS, AB, Canada

WMO: 713930

Lat: 51.1094N

Lon: 114.0003W

Elev: 3547

StdP: 12.91

Time Zone: -7.00 (NAM)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-16.3	-10.9	-22.7	1.8	-16.2	-16.7	2.5	-10.5	26.6	33.7	24.7	33.9	6.2	30	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.2	83.5	59.4	80.1	58.4	77.0	57.5	62.7	76.2	61.0	75.5	59.5	73.2	9.1	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
58.2	82.5	67.9	55.6	75.2	65.4	53.9	70.6	64.1	30.1	75.8	28.7	75.7	27.6	73.0	70.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.4	20.7	18.3	-22.0	89.1	4.0	3.9	-24.9	91.9	-27.3	94.2	-29.6	96.3	-32.5	99.2		
(5)			-22.4	65.9	4.0	2.2	-25.3	67.5	-27.7	68.7	-29.9	70.0	-32.9	71.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	40.2	21.9	18.8	28.3	39.3	49.9	57.0	62.8	61.4	53.3	40.9	28.3	19.1		
	DBStd	19.09	14.30	14.87	13.99	8.92	7.98	5.66	4.96	5.64	8.88	9.28	12.54	13.40		
	HDD50	4841	870	873	673	334	101	9	0	2	66	302	652	959		
	HDD65	9118	1335	1293	1137	771	468	247	102	137	357	746	1101	1424		
	CDD50	1271	0	0	1	13	100	219	396	356	165	20	1	0		
	CDD65	73	0	0	0	0	1	7	33	26	6	0	0	0		
	CDH74	1380	0	0	0	8	64	106	546	462	192	2	0	0		
	CDH80	287	0	0	0	0	3	17	122	109	36	0	0	0		
	WSAvg	7.9	7.1	7.9	7.4	8.4	8.8	9.3	8.1	7.3	7.3	8.3	7.4	7.5		
	PrecAvg	16.7	0.5	0.4	0.7	1.1	2.3	3.4	2.8	2.0	1.8	0.7	0.6	0.5		
(15) Precipitation	PrecMax	24.8	1.1	1.3	2.3	3.1	5.7	7.0	6.3	6.6	5.8	1.9	1.7	1.5		
	PrecMin	11.5	0.0	0.0	0.2	0.0	0.4	0.5	0.4	0.4	0.2	0.0	0.0	0.0		
	PrecStd	2.7	0.3	0.3	0.4	0.7	1.1	1.7	1.4	1.3	1.3	0.4	0.4	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.0	53.8	60.2	73.3	79.7	82.9	87.0	87.7	84.5	70.5	61.3	52.3		
		MCWB	39.8	41.0	43.4	49.7	53.2	60.1	60.6	58.9	56.8	50.4	44.9	36.6		
	2%	DB	47.5	46.6	55.1	65.4	75.3	76.8	83.7	82.8	79.6	64.4	53.0	44.9		
		MCWB	37.0	36.1	40.6	46.5	52.7	56.9	60.2	59.0	56.0	48.0	41.1	35.8		
	5%	DB	44.8	42.7	50.7	59.7	70.8	73.4	79.9	79.2	74.9	60.3	48.5	41.9		
		MCWB	35.6	34.0	38.4	43.3	50.7	55.5	60.0	58.5	54.7	45.9	38.5	33.5		
	10%	DB	41.9	38.6	46.7	54.6	66.5	70.1	76.6	75.4	69.7	56.0	44.5	38.1		
		MCWB	33.9	31.2	36.5	40.5	49.0	53.9	59.4	58.0	52.8	43.6	35.8	31.0		
	0.4%	WB	40.4	40.9	44.4	50.5	55.1	62.1	66.4	64.4	59.3	52.5	45.7	38.4		
		MCDB	50.2	52.9	58.5	71.5	74.9	77.8	76.5	76.4	78.5	67.4	58.5	47.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	37.6	36.8	41.6	47.0	53.2	59.2	63.7	62.1	57.5	49.4	41.9	36.0		
		MCDB	46.4	45.7	53.4	63.9	72.5	72.4	75.8	75.4	75.2	62.3	51.7	45.0		
	5%	WB	36.0	34.1	39.1	44.0	51.5	57.1	61.7	60.4	55.6	46.7	38.9	33.7		
		MCDB	44.1	41.7	49.8	57.8	68.1	69.4	75.8	74.9	71.3	58.2	47.9	41.1		
	10%	WB	34.0	31.6	36.5	41.3	49.9	55.4	60.2	58.9	53.7	44.4	36.3	31.1		
		MCDB	41.7	37.9	46.0	52.7	64.3	66.8	74.0	72.5	68.3	55.6	43.6	37.6		
	5% DB	MDBR	17.8	17.9	18.5	20.2	21.7	20.3	22.2	22.5	22.0	19.8	18.2	17.5		
		MCDBR	20.9	22.6	25.6	29.9	29.5	25.9	29.4	30.7	32.3	26.5	25.1	22.1		
		MCWBR	13.3	13.9	14.3	14.6	12.5	11.0	11.9	11.8	13.0	13.5	14.7	14.8		
		MCWBR	21.0	21.8	24.7	28.9	26.4	22.5	25.0	25.4	28.9	25.5	24.8	21.9		
(40) Clear-Sky Solar Irradiance	taub		0.251	0.268	0.280	0.333	0.341	0.343	0.340	0.355	0.312	0.278	0.253	0.236		
		taud	2.412	2.436	2.477	2.337	2.377	2.407	2.420	2.385	2.475	2.550	2.497	2.409		
	Ebn at Noon		254	280	297	288	290	289	288	277	280	273	254	243		
		Edh at Noon	18	23	28	36	36	35	34	34	28	21	17	16		
(44) All-Sky Solar Radiation	RadAvg		359	670	1076	1479	1752	1872	2055	1679	1184	728	390	279		
	RadStd		25	52	79	107	132	133	123	117	127	78	30	22		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (64 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Nomenclature: See separate page